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The development of a QALY measure for epilepsy: NEWQOL-6D

Brendan Mulhern [a,*], Donna Rowen [a], Ann Jacoby [b], Tony Marson [c], Dee Snape [b], Dyfrig Hughes [d], Nick Latimer [a], Gus A. Baker [c], John E. Brazier [a]

[a] Health Economics and Decision Science, School of Health and Related Research, University of Sheffield, Sheffield, UK
[b] School of Population, Community and Behavioural Sciences, University of Liverpool, UK
[c] Division of Neurological Sciences, University of Liverpool, UK
[d] Centre for Health Economics and Medicines Evaluation, Bangor University, UK

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ABSTRACT

Cost-utility analysis is used to inform the allocation of healthcare resources, using the quality-adjusted life year (QALY) as the outcome measure. We report the development of an epilepsy-specific QALY measure (NEWQOL-6D) derived from the NEWQOL measure of health-related quality of life. Firstly, psychometric and Rasch analyses established the dimension structure of NEWQOL and generated a reduced health state classification system including one item per dimension. Secondly, health states generated by the classification system were valued using Time Trade Off, and the results were modeled to generate a utility score for every health state. A classification system with 6 dimensions (worry about attacks; depression; memory; concentration; stigma; control) was produced, and generalized least squares regression was used to generate utility scores for every health state. This study is the first attempt to derive an epilepsy-specific QALY measure, and the utility values can be used in the economic evaluation of emerging technologies for epilepsy.
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1. Introduction

When assessing emerging treatments and interventions for epilepsy, it is important to investigate both clinical and cost effectiveness. The National Institute for Health and Clinical Excellence (NICE) in the UK requires cost effectiveness to be assessed using quality-adjusted life years (QALYs) as the outcome measure to quantify the benefits in technology appraisal [1]. The QALY is a single measure combining health-related quality of life (HRQL) and the length of time spent in a given health state. Interventions are assessed using cost-per-QALY analysis, a common metric allowing comparisons across all treatments and patient groups.

The 'Q' of the QALY is an index score anchored on a 0 to 1 scale (where 0 is for states equivalent to dead and 1 is for full health). Utility values can be generated using a generic preference-based measures of health, such as EQ-5D [2,3] or SF-6D [4,5]. These QALY measures consist of a health state classification system and a score for every health state defined by the classification system. The EQ-5D classification system includes five dimensions of health and generates 243 health states. A utility score for each health state was generated based on values elicited from a sample of the UK general population.

The validity of generic QALY measures has been questioned in some conditions [6] including epilepsy. Stavem and colleagues [7] found that a number of EQ-5D dimensions discriminated well by seizure status but less well among patients using antiepileptic drugs and patients with neurologic comorbidities. In terms of sensitivity, it has been found that EQ-5D does not capture all of the HRQL dimensions relevant to patients [8] and may be insensitive to seizure control [9]. Generic QALY measures may also fail to capture the full impact of epilepsy on subjective wellbeing [10]. It has also been shown that EQ-5D is not responsive to change in health status across time [8,9], and this is linked to the large ceiling effect observed in EQ-5D responses. The standard recall period for EQ-5D of one day may impact on the sensitivity of the instrument when used for episodic conditions, and can result in higher utility scores (equivalent to better health) [11]. This may impact on the validity of EQ-5D in epilepsy. In response to concerns around using generic measures across a range of disease areas, QALY measures have been developed from existing condition-specific measures of HRQL [12–17]. This involves the reduction of the parent instrument to a shortened health state classification system using psychometric and Rasch analyses. A selection of health states generated by the classification system is valued, and

* Corresponding author at: Health Economics and Decision Science, School of Health and Related Research, University of Sheffield, Regent Court, 30 Regent St, Sheffield, SI 4DA, UK. Fax: +44 114 272 4095.
E-mail address: b.mulhern@sheffield.ac.uk (B. Mulhern).

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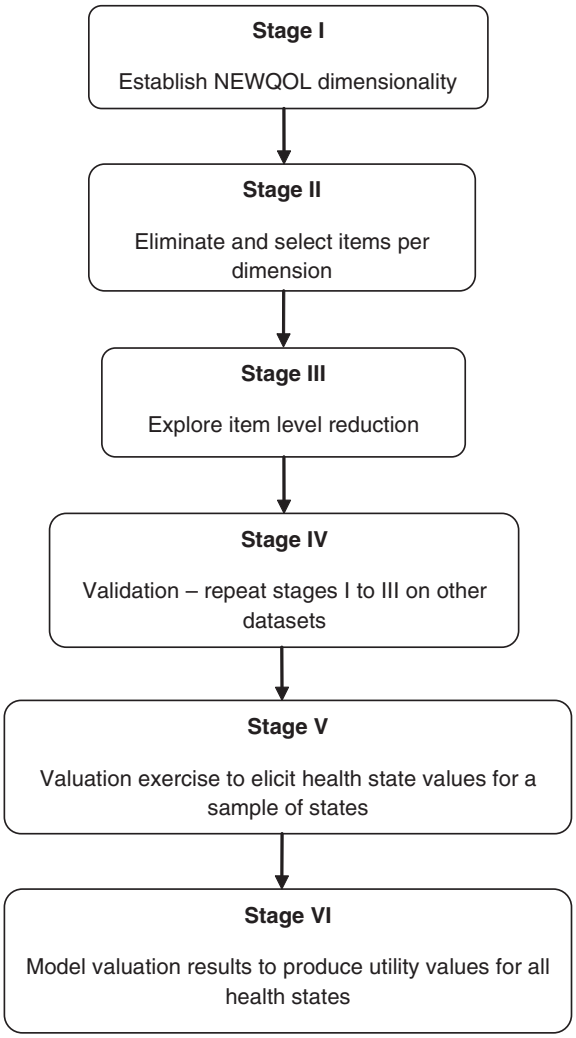

Fig. 1. Flow diagram of the process used to derive a condition-specific health state classification system from a non-preference-based measure.

the values are modeled to estimate a utility value for all possible health states.

This study develops an epilepsy-specific QALY measure from the NEWQOL measure of HRQL in early onset epilepsy. If the measure subsequently proves to be psychometrically valid, then the preference-based index could be used as an alternative to generic measures for the generation of QALYs to inform resource allocation. Following validity testing, the measure may also be used alongside generic instruments to provide decision makers with valuable additional information and potentially improve the applicability of utility scores used for economic evaluations in epilepsy.

2. Measure development process

The development of a QALY measure from the NEWQOL battery followed a six-stage process briefly described below (see Fig. 1). Stages I to IV developed a reduced multidimensional health state classification system from the parent instrument, and this is described in more detail elsewhere [18]. This process is undertaken because NEWQOL has 82 items in total, and this would define millions of states that would be too large for respondents to value. The aim is to produce a classification system that is amenable to valuation but retains the desirable psychometric properties of NEWQOL. This is done by selecting a minimum number of items to represent each important dimension in NEWQOL. Stages V and VI involved the valuation of a sample of states and subsequent modeling to derive utility values for every possible health state.

3. Part 1: health state classification system development (Stages I–IV)

3.1. Methods

3.1.1. Measure

The NEWQOL battery was developed by members of the study team (AJ, GB) and independently validated in the study by Abetz and colleagues [19] to assess HRQL in recently diagnosed epilepsy. NEWQOL includes a range of measures validated for general use across a range of conditions and specifically for epilepsy (see Table 1). A subset of 82 items was considered in the development of the classification system.

3.1.2. Classification system development sample

The sample used to generate the classification system consists of 1611 respondents with newly diagnosed epilepsy. We used baseline data collected for the SANAD study [26,27], an RCT of immediate and deferred antiepileptic drug treatment. Among the SANAD trial population, the mean age was 39 (range 16 to 86), and 54% were female. Overall, 70% reported that their general health was “good” or “better”, and 40% had experienced 10 or more seizures.

3.1.3. Stage I: establish NEWQOL dimensionality

The dimensions from the NEWQOL that are to be included in the health state classification system were established using exploratory factor analysis (EFA) and input from epilepsy clinicians (TM, GB). Exploratory factor analysis establishes the theoretical constructs or

Table 1
Items included in NEWQOL subset (adapted from Abetz et al. [16]).

Scale name	Subscales and items	Scale definition
Worry about seizures scale [20]	2 items	Measures worry about past and possible future seizures using a 4-point scale
Liverpool adverse events profile [21]	19 items	Measures a range of adverse events using a 4-point frequency scale
Hospital anxiety and depression scale [22]	14 items, anxiety (7) and depression (7) subscales	Identifies clinical cases of anxiety and depression using 5-point scales
Social limitations [23]	1 item	Assesses the extent of perceived limitation of social activities on a 4-point scale
Mastery/locus of control scale [24]	7 items	Measures degree of internal vs. external locus of control using a 4-point scale
Stigma of epilepsy scale [20]	3 items	Measures perceived level of stigma associated with epilepsy using a 4-point scale
Impact of seizures scale [23]	12 items	Measures the perceived impact of seizures on a range of life areas using a 5-point scale
AB neuropsychological assessment schedule [25]	24 items, fatigue (8), memory (5), concentration (6), motor (3) and reading (2) subscales	Measures 5 aspects of cognitive function using a 4-point scale

factors underlying a measure and the magnitude of the correlation between each factor and each item (known as factor loading).

3.1.4. Stage II: eliminate and select items per dimension

The first part of stage II is to eliminate items that do not accurately represent the dimension. This was done using Rasch analysis [28], which assesses the performance of individual items in relation to the underlying trait measured by the dimension level model. Rasch analysis was applied to each of the dimensions established at stage I. Items were initially excluded if the response levels were disordered, meaning that individuals could not distinguish between response levels or if they displayed differential item functioning (DIF), which measures if items systematically perform differently across different sample characteristics (gender and age group, split as 16–35 (47%), 36–55 (34%) and 56 or older (19%)). The items were then eliminated if they did not fit the trait measured by the Rasch model. The items were excluded sequentially, and the Rasch model refitted after each exclusion. The data were split randomly into two halves and the analysis was repeated on each subset. The second part of stage II is to select the most appropriate remaining items for each dimension of the classification system. Rasch statistics that assess the magnitude to which the remaining items measure the full severity of the underlying trait guided the selection process. Item fit to the dimension level Rasch model and clinical input also informed the selection process.

3.1.5. Stage III: validation of items selected

As part of the development of a condition-specific QALY measure, Brazier et al. [29] recommend validating the results of stage II on an external dataset including the condition-specific measure or on a subset of the dataset used to develop the classification system if possible. In this study, Rasch analysis was carried out on one subset of the data (data half 1), and validated on a second subset of the data (data half 2). The two data subsets were randomly generated. Subset analysis was possible as the optimum number required for the Rasch analysis is 500 [30], and the large dataset used to develop the health state classification system allowed for two data subsets to be randomly generated.

3.1.6. Stage IV: exploration of item level reduction

Stage IV assessed whether the number of response levels used for each item selected at stages II and III could be reduced. Rasch analysis was used to investigate the ordering of responses, and the distribution of responses across each option was also assessed.

3.2. Results

3.2.1. Stage I: establish NEWQOL dimensionality

A six-dimension structure, with dimensions defined as worry about seizures, depression, memory cognition, control and stigma was used as the basis to develop the classification system (Table 2).

3.2.2. Stages II and III: eliminate and select items per dimension and validation of results

Table 2 displays the item elimination and selection process, with results combined for both the development (half 1) and validation (half 2) data subsets (stage III of the development process). At this stage, 18 items were eliminated because the response levels of 2 items were disordered, 11 displayed DIF, and 5 did not fit the dimension level Rasch model. For the worry dimension, the item assessing 'worry about future seizures' was chosen, as this is an important issue for patients with epilepsy. For the depression dimension the item 'depression' was used, as it directly assesses the dimension concept. The only item remaining for the memory dimension was 'memory problems'. For the cognition dimension, 'I can't concentrate for more than a short period of time' was chosen, as it displays good severity coverage and assesses a cognition-related issue that is more prevalent than the concepts assessed by the other remaining items. For the control dimension, 'I have little control over things that happen to me' covers more of the severe end of the severity spectrum and displays larger spread than the other remaining items. For stigma, the item 'I feel some people treat me like an inferior person' displayed slightly better severity coverage and overall item fit.

3.2.3. Stage IV: exploration of item level reduction

Analysis suggested that the responses to the items are spread sufficiently well across the response options. Therefore, it was decided to use the original four-level categories for each NEWQOL dimension.

3.2.4. Final classification system

The final classification system (named NEWQOL-6D) is displayed in Fig. 2. Each dimension has four health state levels, and the classification system generates a possible 4096 (i.e., 4^6) health states. The cognition dimension was renamed 'concentration' to reflect the item chosen. Health states are generated using one severity level of each dimension.

4. Part 2: deriving utility values

4.1. Methods

4.1.1. Stage V: valuation study

Stage V involved the valuation of a selection of health states generated by the classification system using the preference elicitation technique Time Trade Off (TTO). Due to the large number of states produced by the classification system, it was not feasible to value all states, so a sample of 50 states (including the best (111111) and worst states (444444)) was selected using an orthogonal array. The 50 states were allocated to nine blocs of eight states (where each bloc included the worst state). The allocation process was done by summing the dimension levels to give an overall indicator of severity and allocating a mixture of mild, moderate and severe states to each bloc. The states were valued by a representative sample of the UK population who completed one bloc each. One bloc was administered to more respondents to allow for comparability with TTO valuations provided by an epilepsy patient sample (who have valued the same bloc as part of a related study). Ethical approval for the study was granted by the University of Sheffield School of Health and Related Research ethics committee, and informed consent was obtained from each respondent.

The TTO study used face-to-face interviews carried out in the respondent's home. At the start of the interview, respondents completed the EQ-5D and the NEWQOL classification system to ensure that they were familiar with the dimensions and levels included in the health states. Respondents were informed that the health states were related to epilepsy. First, respondents ranked the 8 health states that they were valuing in addition to states defined as 'full heath' and 'dead' in order from best to worst. Second, respondents valued the health states following the TTO protocol used to value EQ-5D [3]. Respondents were asked whether they would prefer to live in the health state for 10 years or to die immediately. This established whether the health state was perceived as better than, worse than, or equal to being dead.

For states better than dead, respondents chose between a) living in the health state for 10 years or b) living in full health for x years (where $x \leq 10$). The duration of full health (x) was varied until indifference between the scenarios was reached. The value for the health state is x/10. For states worse than dead, respondents chose between a) the selected health state for w years (where $w = 10 - x$) followed by full health for x years or b) immediate death. Years in full health

Table 2
Dimensions and items used to generate the classification system, and the item exclusion and selection process (stages I/II and IV).

Dimension	Item	Item performance (development data subset)					Item performance (validation data subset)					Item selected
		Responses ordered?	No DIF[a]?	Good fit?	Range[b]	Spread[c]	Responses ordered?	No DIF?	Good fit?	Range[b]	Spread[c]	
Worry	How worried are you about the attacks you have had?	√	√	√	−3.36 to 2.96	0.03 to 0.97	√	√	√	−3.07 to 3.38	0.03 to 0.96	
	How worried are you that you might have another attack?	√	√	√	−3.78 to 2.96	0.05 to 0.98	√	√	√	−3.34 to 2.88	0.05 to 0.97	√
Depression	Depression	√	√	√	−2.04 to 0.64	0.35 to 0.88	√	√	√	−2.20 to 0.44	0.39 to 0.90	√
	I still enjoy the things I used to enjoy	✗	✗	√	N/A	N/A	✗	✗	√	N/A	N/A	
	I can laugh and see the funny side of things	√	√	√	−0.71 to 2.63	0.07 to 0.67	√	√	√	−0.93 to 2.87	0.05 to 0.72	
	I feel cheerful	√	√	√	−1.53 to 2.20	0.10 to 0.82	√	✗	√	N/A	N/A	
	I look forward with enjoyment to things	√	✗	√	N/A	N/A	√	✗	√	N/A	N/A	
	I can enjoy a good book or radio or TV program	✗	√	✗	N/A	N/A	✗	√	✗	N/A	N/A	
Memory	Memory problems	√	√	√	−2.37 to 0.58	0.36 to 0.91	√	√	√	−2.36 to 0.63	0.35 to 0.91	√
	I have difficulties remembering the names of people	√	✗	√	N/A	N/A	√	✗	√	N/A	N/A	
	I forget all kinds of things, for example an appointment or where I put an object	√	√	√	−1.78 to 0.80	0.31 to 0.86	√	√	✗	N/A	N/A	
	I forget things people have said to me	√	√	✗	N/A	N/A	√	√	✗	N/A	N/A	
	I get confused and forget what I was doing	√	✗	√	N/A	N/A	√	√	√	−0.64 to 1.73	0.15 to 0.65	
Cognition	My mind does not work as fast as it should	√	✗	√	N/A	N/A	√	✗	√	N/A	N/A	
	I have difficulties in following a book or film	√	√	√	−0.39 to 1.55	0.18 to 0.60	√	✗	√	N/A	N/A	
	I have problems finding the correct word	√	√	✗	N/A	N/A	√	✗	√	N/A	N/A	
	My thinking has slowed down	√	✗	√	N/A	N/A	√	✗	√	N/A	N/A	
	I have difficulty concentrating on the things that I am doing.	√	√	✗	N/A	N/A	√	√	✗	N/A	N/A	
	I have problems understanding what I read	√	√	√	0.31 to 2.51	0.07 to 0.47	√	√	√	0.24 to 1.85	0.14 to 0.44	
	I can't concentrate for more than a short period of time	√	√	√	−1.42 to 1.78	0.14 to 0.80	√	√	√	−1.49 to 1.56	0.17 to 0.82	√
	I get distracted easily	√	√	√	−1.65 to 1.61	0.17 to 0.84	√	√	√	−1.75 to 1.39	0.20 to 0.85	
	I sometimes stutter or am unable to find the correct words	√	√	✗	N/A	N/A	√	√	√	−1.28 to 1.06	0.26 to 0.78	
	I feel I react too slowly to things that are said to me	√	√	√	−1.05 to 2.17	0.10 to 0.74	√	√	√	−1.25 to 1.56	0.17 to 0.78	
Control	There is really no way I can solve some of the problems I have	√	√	√	−1.96 to 0.74	0.32 to 0.88	√	√	√	−2.27 to 0.76	0.32 to 0.91	
	I sometimes feel that I am pushed around in my life	√	✗	√	N/A	N/A	√	✗	√	N/A	N/A	
	I have little control over things that happen to me	√	√	√	−1.86 to 1.57	0.17 to 0.87	√	√	√	−2.11 to 1.35	0.21 to 0.89	√
	I often feel helpless in dealing with the problems of life	√	✗	√	N/A	N/A	√	√	√	−2.05 to 1.49	0.18 to 0.89	
	There is little I can do to change many of the important things in my life	√	✗	√	N/A	N/A	√	✗	√	N/A	N/A	
Stigma	I feel some people are uncomfortable with me	√	√	✗	N/A	N/A	√	√	✗	N/A	N/A	
	I feel some people treat me like an inferior person	√	√	√	−1.54 to 1.38	0.20 to 0.82	√	√	√	−1.23 to 1.24	0.22 to 0.77	√
	I feel some people would prefer to avoid me	√	√	√	−1.40 to 1.39	0.20 to 0.80	√	√	√	−1.25 to 0.69	0.34 to 0.78	

[a] DIF: Differential Item Functioning.
[b] The larger the item range, the better the severity coverage.
[c] Spread value is between 0 and 1 where a larger range indicates better severity coverage.

(x) and years in the selected health state (w) were varied until the point of indifference between the scenarios was reached. The value for the state is x/10.

4.1.2. Stage VI: modeling valuation study results to produce utility values

The TTO results were modeled to estimate utility values for every health state defined by the classification system using individual and mean level multivariate regression models. The models estimate disutility preference weights for every severity level of each dimension using level 1 (no problem) as the baseline. Full health is anchored at 1, with classification system-specific full health equivalent to 1 minus the constant term. The utility value for each health state is calculated by subtracting the constant term and the disutility value for each dimension from 1. Ordinary least squares (OLS), Random Effects Generalized Least Squares (RE GLS), and Tobit models were estimated on individual level data. Tobit models were estimated to take account of the bounded nature of the data, as a large proportion of TTO observations were at the maximum level of 1. Models were estimated including

Worry about attacks
You are not worried at all that you might have another epileptic attack
You are a little worried that you might have another epileptic attack
You are fairly worried that you might have another epileptic attack
You are very worried that you might have another epileptic attack

Depression
You never have problems with depression
You rarely have problems with depression
You sometimes have problems with depression
You always or often have problems with depression

Memory
You never have problems with your memory
You rarely have problems with your memory
You sometimes have problems with your memory
You always or often have problems with your memory

Concentration
You have no problem concentrating for more than a short period of time
You have mild problems concentrating for more than a short period of time
You have moderate problems concentrating for more than a short period of time
You have serious problems concentrating for more than a short period of time

Control
You feel that you have complete control over things that happen to you
You feel that you have some control over things that happen to you
You feel that you have little control over things that happen to you
You feel that you have no control over things that happen to you

Stigma
You do not feel that people treat you like an inferior person
You feel that some people maybe treat you like an inferior person
You feel that some people probably treat you like an inferior person
You feel that some people definitely treat you like an inferior person

Fig. 2. Final health state classification system.

an N4 term, which adds a utility decrement if any dimension is at the most severe level. Ordinary least squares mean level models, where the data consist of the mean value for each state, were also estimated.

The number of inconsistent and significant coefficients, root mean squared error (RMSE) and the mean absolute error (MAE) of health state predictions were used to investigate model performance. When coefficients between adjacent levels of a dimension are inconsistent, this implies that when health worsens, health utility, as measured by the dimension, increases. Inconsistent dimensions were combined to achieve consistency. Mean absolute error indicates how large prediction errors are at the health state level. To assess the precision of the estimations, plots of actual and predicted mean TTO scores were investigated.

Table 3
Characteristics of respondents.

	Respondents	South Yorkshire[a]	England[a]
Mean age (s.d.)	46.18 (15.89)	n/a	n/a
Age categories			
18–40	39.0%	41.2%	41.6%
41–65	51.7%	39.1%	39.1%
Over 65	9.3%	19.7%	19.3%
Female	63.0%	51.2%	51.3%
Married/partner	71.2%	n/a	n/a
Employed/self employed	57.1%	56.1%	60.9%
Unemployed	0.5%	4.1%	3.4%
Long-term sick	1.1%	7.7%	5.3%
Full-time student	9.2%	7.5%	7.3%
Retired	18.5%	14.4%	13.5%
Secondary school highest level of education	31.0%	n/a	n/a
EQ-5D score	0.88 (0.19)		0.86 (0.23)[b]

nb:
[a] Statistics from the South Yorkshire Health Authority and Census for England 2001.
[b] Statistic from the Measurement and Valuation of Health (MVH) study in 1993 (Kind et al., 1999).

Table 4
Descriptive statistics of observed TTO for NEWQoL-6D.

Health state	Mean	s.d.	Median	Count
332112	0.953	0.07	1	20
212223	0.945	0.07	1	20
111111	0.942	0.18	1	60
122233	0.940	0.09	1	18
211214	0.936	0.11	1	19
114224	0.927	0.09	0.938	20
213211	0.922	0.14	1	18
133322	0.914	0.11	0.950	21
122211	0.913	0.19	1	20
121242	0.907	0.12	0.950	21
121321	0.906	0.15	1	20
222321	0.900	0.15	1	21
322134	0.895	0.12	0.925	21
311323	0.881	0.22	0.925	21
233121	0.870	0.18	1	21
111133	0.859	0.30	1	21
241232	0.852	0.19	0.925	21
321121	0.846	0.20	0.925	19
113132	0.827	0.22	0.925	60
133344	0.825	0.25	0.938	20
313411	0.817	0.27	0.925	21
231122	0.813	0.21	0.825	19
224312	0.813	0.28	0.925	60
321412	0.808	0.27	0.925	21
434231	0.746	0.18	0.763	20
132313	0.793	0.26	0.900	19
223333	0.793	0.21	0.850	60
212332	0.786	0.25	0.813	20
234113	0.775	0.27	0.913	20
143113	0.771	0.29	0.875	62
313332	0.769	0.28	0.900	21
332222	0.766	0.26	0.825	60
333231	0.757	0.21	0.800	19
342314	0.671	0.37	0.763	20
441331	0.755	0.21	0.763	20
431313	0.710	0.37	0.900	20
242141	0.749	0.27	0.800	21
132431	0.748	0.23	0.700	21
231434	0.741	0.21	0.800	19
331243	0.725	0.26	0.800	21
324133	0.722	0.27	0.800	19
423212	0.798	0.25	0.825	21
412142	0.653	0.35	0.675	19
343223	0.708	0.29	0.775	21
421423	0.717	0.23	0.65	20
144422	0.640	0.34	0.700	59
314341	0.639	0.32	0.600	21
223443	0.624	0.33	0.600	19
423124	0.749	0.23	0.825	21
444444	0.366	0.36	0.375	182

4.2. Results

4.2.1. Stage V: valuation study

4.2.1.1. State selection. The orthogonal array produced a sample of 49 states, to which the worst possible health state produced by the classification system was added. The 50 states were divided into 7 blocks of 8 states (including 7 unique states plus the worst state), and each participant valued one block.

4.2.1.2. Valuation study sample. Overall, 184 interviews were carried out, and two respondents were excluded for valuing all states identically and less than one. Table 3 displays the sample characteristics.

4.2.1.3. Valuation study results. Descriptive statistics for observed TTO scores are shown in Table 4, and the distribution is shown in Fig. 3. Mean observed values range from 0.953 (sd = 0.07) to 0.366

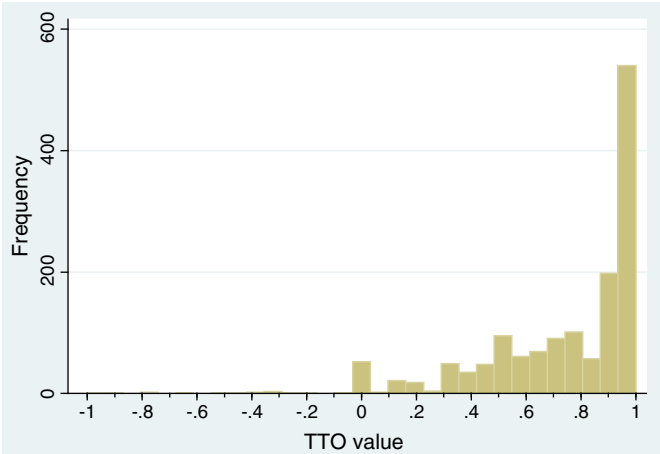

Fig. 3. Histogram of observed TTO values for NEWQoL-6D.

(sd = 0.36). All except two states, (332112 and 212223), had mean values below that of the 'best' state (111111). The worst state (444444) has the lowest mean value overall.

4.2.2. Stage VI: modeling valuation study results to produce utility values

The RE GLS model displayed the best fit in terms of the model indicators used (Table 5 displays the GLS regression models used for NEWQOL-6D). The initial RE GLS model (model 1 in Table 5) has one inconsistency (where the coefficient for level 2 of the depression dimension is inconsistent, meaning that an increase in depression severity leads to an increase rather than a decrease in utility). This level was collapsed to generate RE GLS model 2. This model was used to derive utility values for NEWQOL-6D. Model 2 performs well across all indicators and has the highest number of significant coefficients across all of the models fitted. Utility values for each health state are calculated as 1 minus the constant term minus the disutility for each dimension level (so for example state 223443 has a value of 1 − 0.043 − 0.010 − 0 − 0.035 − 0.092 − 0.125 − 0.045 = 0.650) Fig. 4 plots the predicted and observed TTO values for this model. Utility estimates for model 2 range from 0.341 to 0.957, and this covers the full range of the observed values.

Table 5
Preliminary NEWQoL-6D regression models.

	1. RE GLS	2. RE GLS (consistent)
Worry 2	0.015	0.010
Worry 3	0.048*	0.054*
Worry 4	0.141*	0.132*
Depression 2	−0.021	0
Depression 3	0.021	0.029**
Depression 4	0.096*	0.116*
Memory 2	0.027***	0.031***
Memory 3	0.026***	0.035**
Memory 4	0.081*	0.093*
Concentration 2	0.006	0.004
Concentration 3	0.025*	0.024
Concentration 4	0.085*	0.092*
Control 2	0.015	0.020
Control 3	0.045*	0.052*
Control 4	0.114*	0.125*
Stigma 2	0.025	0.027***
Stigma 3	0.048*	0.045*
Stigma 4	0.064*	0.058*
Constant	0.042	0.043
N4 decrement		
Obs	1453	1453
N	182	182
Adj. R-squared	0.272	0.293
No. of inconsistencies	1	0
No. of sig. variables	12	13
RMSE	0.050	0.052
MAE	0.063	0.065
% with MAE > 0.05	44	44
% with MAE > 0.10	12	14

* Significant at 1%.
** Significant at 5%.
*** Significant at 10%.

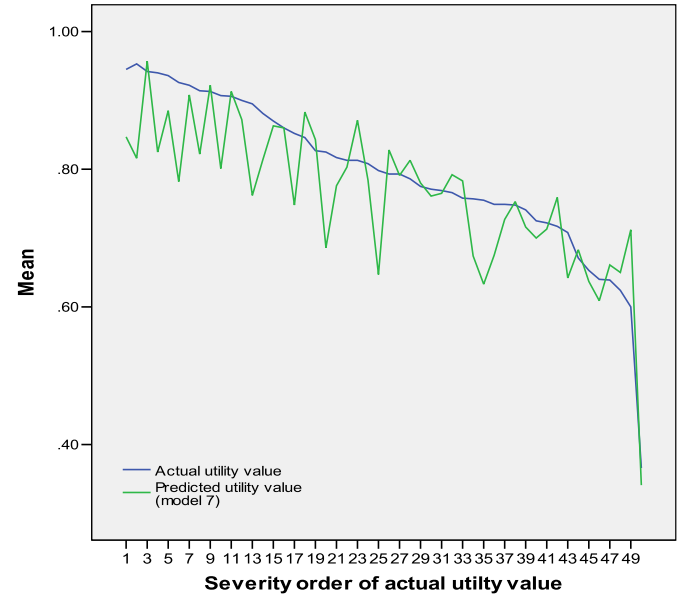

Fig. 4. Observed and predicted values for NEWQoL-6D using model 7 (ordered by mean observed TTO value).

5. Discussion

This article describes the development of an epilepsy-specific QALY measure, NEWQOL-6D, from the NEWQOL battery measure. The derivation of the QALY measure follows a six-stage process [29] that develops a reduced classification system, values a range of health states and models these values to generate utility values for all NEWQOL-6D health states. The NEWQOL-6D enables a utility value that can be used to generate QALYs for economic evaluation to be generated each time a respondent completes NEWQOL and can be applied to existing datasets.

It should be noted, however, that further testing of the psychometric validity of NEWQOL-6D is required before utility values derived from the classification system can be used to inform economic evaluations in epilepsy. Until the results of this analysis are available, utility values generated by NEWQOL-6D cannot be used with confidence. As the original NEWQOL battery includes a large number of items and standalone measures, there is also a concern that missing data may impact on the generation of utility scores (which requires all six dimensions to be completed). Phase 2 of the NEWQOL-6D development process will examine the acceptability (i.e., missing data rates), validity and responsiveness of the instrument in comparison to EQ-5D. This will be done using an epilepsy trial dataset. To further investigate the NEWQOL-6D preference-based index, we will also examine whether health state utility values for NEWQOL-6D differ when elicited from the general population, as described in this paper, or from people with epilepsy.

The process used to develop the classification system builds on previous work using psychometric and Rasch analyses to generate classification systems amenable to valuation [12,17]. The analyses inform the selection of items for inclusion by quantifying item performance. Clinical input, during the selection process, maximizes face

validity and enables the most relevant item to be selected. This study is the first attempt to derive a classification system from a battery measure that includes a variety of validated instruments developed for epilepsy (ABNAS and AEP) and also a widely used measure (HADS) that is an accepted indicator of depression and anxiety across a range of conditions [31]. The methodology applied in the valuation study was successfully used to value a range of condition-specific health state classification systems [13–16] and also to derive utility values for EQ-5D [3], and this allows for a level of comparability across different QALY measures.

The use of generic QALY measures, such as EQ-5D in epilepsy, has been criticized as the classification system does not fully reflect the impact of the epilepsy on HRQL, and the utility index is not responsive to changes in epilepsy patient groups over time [7–10]. The one-day recall period of the EQ-5D may also mean that the classification system is insensitive in episodic conditions [11]. This means that QALY estimates produced using generic measures may not be accurate. NEWQOL-6D may address some of these concerns by providing utility values that reflect epilepsy-specific HRQL in terms of the dimensions included in the instrument and also by using a one-week recall period. However, this cannot be established until there is evidence regarding the validity of NEWQOL-6D.

To generate QALYs for agencies such as NICE who recommend the use of EQ-5D [1], it may be beneficial to include both generic and condition-specific QALY measures, as unlike generic instruments, NEWQOL-6D may not capture all comorbidities or adverse effects related to epilepsy treatment. The advantage of using the epilepsy-specific NEWQOL-6D is that it is focused on dimensions that are most relevant for assessing the health of epilepsy patients and is useful for informing comparisons of interventions for epilepsy. However, for resource allocation across all interventions and patient groups, the use of a generic measure is preferred.

There are a number of limitations relating to both the development of the classification system and the valuation study. The classification system may not cover all of the relevant epilepsy-specific HRQL dimensions. For example, social and activity restrictions due to epilepsy are not included in the classification system, and this is because the relevant items included in the parent instrument are not amenable to preference valuation, as they assess transition in activity level between time points. The factor and Rasch analyses may also exclude areas of HRQL included in the original NEWQOL as the item reduction process used to generate a classification system amenable to valuation is necessarily stringent. However, clinical input into the final dimension structure helped to ensure that a full range of relevant dimensions was included. The generalizability of NEWQOL-6D to all epilepsy populations may be limited, as NEWQOL was specifically developed for use in early-onset epilepsy. However, further psychometric analysis of the NEWQOL instruments across a range of epilepsy populations will allow this issue to be investigated in more detail. Furthermore, we developed the health state classification system using a random subset of the sample and validated the results on another subset of the data, but we were not able to validate the classification system on an external dataset including NEWQOL. This may be possible in future studies using NEWQOL as a measure of HRQL in epilepsy research. The small number of states used in the valuation study is also a limitation, particularly in comparison to the total number of health states produced by NEWQOL-6D. The sample size used for the valuation study is small in comparison to other condition-specific QALY measures and the EQ-5D valuation studies due to funding limitations. A larger sample size with more states may generate a more precise estimate of the utility related to NEWQOL-6D health states.

It should also be noted that five of the eighteen health state parameters used to calculate the disutility of each NEWQOL health state are insignificant. This is a common issue which is also found for selected parameters of both EQ-5D [3] and SF-6D [4,5]. It is possible to merge nonsignificant parameters with adjacent levels to derive a model where all of the disutility values are significant. However, this has implications for the sensitivity of the measure to change, as it limits the ability to distinguish between severity levels. Therefore, as we believe it is important to show a decrease in utility at each level, nonsignificant parameters are included in the model, and only inconsistent parameter values are merged with adjacent levels.

We have described the first attempt to derive an epilepsy-specific QALY measure from an existing parent instrument of HRQL that can generate QALYs for economic evaluation. Patient responses to the NEWQOL can be converted into NEWQOL-6D utility scores to generate QALYs. Following psychometric validation of the instrument, it is possible that the utility weights could be used in economic evaluations to complement those derived from generic measures such as the EQ-5D. If NEWQOL-6D proves valid, the preference weights may also be used when generic utility scores are not available and may help to address some of the concerns around using generic QALY measures in epilepsy.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at doi:10.1016/j.yebeh.2012.02.025.

References

[1] National Institute of Health and Clinical Excellence. Guide to the methods of technology appraisal. London: NICE; 2008.
[2] Brooks R. EuroQol: the current state of play. Health Policy 1996;37(1):53–72.
[3] Dolan P. Modeling valuations for EuroQol health states. Med Care 1997;35(11):1095–108.
[4] Brazier JE, Roberts J. Estimating a preference-based index from the SF-12. Med Care 2004;42(9):851–9.
[5] Brazier J, Roberts J, Deverill M. The estimation of a preference-based measure of health from the SF 36. J Health Econ 2002;21(2):271–92.
[6] Brazier J, Longworth L. NICE DSU technical support document 8: an introduction to the measurement and valuation of health for NICE submissions. London: NICE Decision Support Unit; 2011.
[7] Stavem K, Bjornaes H, Lossius MI. Properties of the 15D and EQ-5D utility measures in a community sample of people with epilepsy. Epilepsy Res 2001;44:179–89.
[8] Selai CE, Elstner K, Trimble MR. Quality of life pre and post epilepsy surgery. Epilepsy Res 2000;38:67–74.
[9] Langfitt JT, Vickrey BG, McDermott MP, et al. Validity and responsiveness of generic preference-based HRQOL instruments in chronic epilepsy. Qual Life Res 2006;15:899–914.
[10] Bockerman P, Johansson E, Saarni SI. Do established health-related quality-of-life measures adequately capture the impact of chronic conditions on subjective well-being. Health Policy 2011;100:91–5.
[11] Bansback N, Sun H, Guh DP, et al. Impact of the recall period on measuring health utilities for acute events. Health Econ 2008;17:1413–9.
[12] Mulhern B, Smith SC, Rowen D, et al. Improving the measurement of QALYs in dementia: Developing patient and carer reported health state classification systems using Rasch analysis. Value Health 2012;15(2):323–33.
[13] Rowen D, Brazier JE, Young T, Gaugris S, Craig BM, King MT, Velikova G. Deriving a preference-based measure for cancer using the EORTC QLQ-C30. Value Health 2011;14:721–31.
[14] Rowen D, Mulhern B, Banerjee S, van Hout B, Knapp M, Smith S, Lamping D, Brazier J. Estimating preference based single index measures for dementia using DEMQOL and DEMQOL-Proxy. Value Health 2012;15(2):346–56.
[15] Yang Y, Brazier J, Tsuchiya A, Coyne K. Estimating a preference-based index from the overactive bladder questionnaire. Value Health 2008;12:59–66.
[16] Yang Y, Brazier JE, Tsuchiya A, Young T. Estimating a preference based index for a 5-dimensional health state classification for asthma derived from the asthma quality of life questionnaire. Med Decis Making 2011;31:281–91.
[17] Young TA, Yang Y, Brazier JE, Tsuchiya A, Coyne K. The first stage of developing preference based measures: constructing a health state classification using Rasch analysis. Qual Life Res 2009;18(2):253–65.

[18] Mulhern B, Rowen D, Brazier JE, et al. Developing a health state classification system from NEWQOL for epilepsy using classical psychometric techniques and Rasch analysis: a technical report. HEDS Discuss Pap 2010;10(16):1–31.
[19] Abetz L, Jacoby A, Baker GA, McNulty P. Patient-based assessments of quality of life in newly diagnosed epilepsy patients: validation of the NEWQOL. Epilepsia 2000;41(9):1119–28.
[20] Jacoby A. Felt versus enacted stigma: a concept revisited: evidence from a study of people with epilepsy in remission. Soc Sci Med 1994;38:269–74.
[21] Baker GA, Jacoby A, Francis P, Chadwick DW. The Liverpool adverse drug events profile. Epilepsia 1995;36:S59.
[22] Zigmond AS, Snaith RP. The hospital anxiety and depression scale. Acta Psychiatr Scand 1983;67:361–70.
[23] Jacoby A, Johnson AL, Chadwick DW. Psychosocial outcomes of antiepileptic drug discontinuation. Epilepsia 1992;33:123–31.
[24] Pearlin L, Schooler C. The structure of coping. J Health Soc Behav 1978;19:2–21.
[25] Aldenkamp AP, van Meel HF, Baker GA, Brooks J, Hendriks MPH. The A–B neuropsychological assessment schedule (ABNAS): the relationship between patient-perceived drug related cognitive impairment and results of neuropsychological tests. Seizure 2002;11(4):231–7.
[26] Marson AG, Al-Kharusi AM, Alwaidh M, et al. The SANAD study of effectiveness of carbamazepine, gabapentin, lamotrigine, oxcarbazeprine, or topiramate for treatment of partial epilepsy: an unblended randomised controlled trial. Lancet 2007;369:1000–15.
[27] Marson AG, Appleton R, Baker GA, et al. A randomised controlled trial examining the longer-term outcomes of standard versus new antiepileptic drugs. The SANAD trial. Health Technol Assess 2007;11(37):1–145.
[28] Rasch G. Probabilistic models for some intelligence and attainment tests. Chicago: University of Chicago Press; 1960.
[29] Brazier J, Rowen D, Mavranezouli I, et al. Developing and testing methods for deriving preference-based measures of health from condition specific measures (and other patient based measures of outcome). Health Technol Assess in press.
[30] Linacre JM. Investigating rating scale category utility. J Outcome Meas 1999;3(2):103–22.
[31] Bjelland I, Dahl AA, Haug T, Neckelmann D. The validity of the hospital anxiety and depression scale. An updated literature review. J Psychosom Res 2002;52(2):69–77.